

Amazon S3 (Simple Storage Service) - Service Overview

Introduction

Amazon Simple Storage Service (S3) is an object storage service offering industry-leading scalability, data availability, security, and performance. S3 stores data as objects within buckets, and can store virtually unlimited amounts of data, from a few bytes to multiple terabytes per object.

Buckets and Objects

A bucket is a container for objects. Bucket names must be globally unique across all of AWS. Each object consists of data, a key (its unique identifier within the bucket), and metadata. Objects can range in size from 0 bytes to 5 terabytes.

Storage Classes

S3 offers storage classes optimized for different access patterns and cost profiles:

- S3 Standard: for frequently accessed data, offering low latency and high throughput.
- S3 Intelligent-Tiering: automatically moves objects between access tiers based on changing access patterns, with no retrieval fees.
- S3 Standard-IA (Infrequent Access): lower storage cost for data accessed less frequently, but retrieval fees apply.
- S3 One Zone-IA: similar to Standard-IA but stored in a single Availability Zone, at lower cost.
- S3 Glacier Instant Retrieval, Flexible Retrieval, and Deep Archive: designed for long-term archival, with retrieval times ranging from milliseconds to 12+ hours, at progressively lower storage cost.

Durability and Availability

S3 Standard is designed for 99.999999999% (11 nines) durability and 99.99% availability over a given year, achieved by automatically storing data across a minimum of three Availability Zones.

Versioning

When enabled, S3 Versioning keeps multiple variants of an object in the same bucket, protecting against accidental deletion or overwrite. Deleting an object in a versioned bucket inserts a delete marker rather than permanently removing the object, and prior versions remain retrievable.

Access Control

S3 access can be controlled through:

- Bucket policies: JSON-based resource policies attached directly to the bucket.
- IAM policies: attached to users, groups, or roles, granting or denying S3 actions.

- Access Control Lists (ACLs): a legacy access control mechanism, generally discouraged in favor of IAM and bucket policies.
- S3 Block Public Access: an account or bucket-level setting that overrides other permissions to prevent accidental public exposure.

Encryption

S3 supports server-side encryption (SSE-S3, SSE-KMS, SSE-C) and client-side encryption. SSE-KMS additionally provides an audit trail of key usage via AWS CloudTrail.

Presigned URLs

A presigned URL grants temporary access to a private object without requiring the requester to have AWS credentials. The URL is generated using the credentials of a principal who has permission to access the object, and includes an expiration time set by the generator.

Lifecycle Policies

An S3 Lifecycle configuration defines rules that automatically transition objects to a cheaper storage class, or expire (delete) them, after a specified number of days. For example, a rule might transition objects from S3 Standard to S3 Standard-IA after 30 days, then to S3 Glacier Flexible Retrieval after 90 days, and finally expire them entirely after 365 days. This is commonly used for log files and audit trails that are accessed frequently when new, rarely accessed as they age, and eventually no longer need to be retained at all.

Cross-Region Replication

S3 Cross-Region Replication (CRR) automatically and asynchronously copies objects from a source bucket in one AWS region to a destination bucket in a different region, for disaster recovery, compliance requirements around data residency, or reducing latency for users accessing the data from a different geography. Both source and destination buckets must have versioning enabled for CRR to function.

Common Use Cases

- Static website hosting
- Data lake storage for analytics workloads
- Backup and disaster recovery
- Storing application assets such as images, documents, and logs
- Hosting Docker build contexts, Terraform state files (with locking via DynamoDB), or CI/CD

artifacts

Best Practices

- Enable versioning and MFA delete for buckets containing critical data.
- Use S3 Lifecycle policies to automatically transition or expire objects based on age.
- Enable S3 Block Public Access at the account level unless a bucket is explicitly intended to be public.
- Use least-privilege bucket policies scoped to specific prefixes rather than granting access to an entire bucket.